

Notes: Piccolo and Shapiro (RFS, 2022)
Credit Ratings and Market Information

Contents

1 Summary	3
2 Big picture	3
3 Incremental contribution of the paper	4
4 Model setup	4
4.1 The ratings process	4
4.2 The credit-risk market	5
4.3 CRA type and reputation	5
5 Timing of the model	5
6 The CDS market equilibrium	6
6.1 Trading strategy and price signal	6
6.2 Strategic substitute from the speculator's perspective	6
7 The rating game	6
7.1 CRA payoff	6
7.2 Strategic complement from the CRA's perspective	7
8 Joint equilibrium	7
9 Comparative statics	7
10 Real effects and welfare	8
11 Extensions	8
11.1 Linked primary and secondary markets	8
11.2 Multiple rating agencies	8
11.3 Inflated ratings	9
11.4 Project outcomes, other CRAs, and media	9
12 Empirical implications	9

13 How to use the paper	10
14 Limitations and caveats	10
15 Original paper extracts	11
15.1 Title, model environment, and timing	11
15.2 Strategic interaction between the CRA and the CDS market	12
15.3 Equilibrium and comparative statics	13
15.4 Real effects and linked markets	14
15.5 Multiple CRAs and rating inflation	14
15.6 Other sources of discipline and appendix figures	15
16 Notes-created tables	17
16.1 Key objects in the model	17
16.2 Main results	18
16.3 Policy interpretation	19
17 Short summary	19
18 Conclusion	20
18.1 Core conclusion	20
18.2 Economic intuition	20
18.3 Incremental contribution revisited	20
18.4 Takeaway	20

1 Summary

Piccolo and Shapiro study how information in credit-risk markets can discipline credit rating agencies (CRAs). The core idea is that a CRA's rating is followed by a market for credit risk, such as a CDS market or secondary bond market. Prices in that market reveal information about whether the rating was likely correct. This makes rating errors more transparent and strengthens the CRA's reputational incentives to acquire accurate information before issuing the rating.

The paper's central mechanism is a two-way strategic interaction. From the perspective of the CRA, market informativeness and rating accuracy are **strategic complements**: if the CDS market is more informative, a bad rating is more likely to be exposed, so the CRA has stronger incentives to be accurate. From the perspective of the speculator in the CDS market, market informativeness and rating accuracy are **strategic substitutes**: if ratings are already very accurate, there is less mispricing for the speculator to exploit, so the speculator has weaker incentives to acquire information.

These two forces lead to a unique equilibrium in which rating accuracy and market informativeness are jointly determined. In this equilibrium, deeper and more informative credit-risk markets can increase rating accuracy through the crowding-in channel. At the same time, direct regulatory discipline on the CRA can crowd out market discipline because more accurate ratings reduce the speculator's incentive to become informed.

A central welfare point is that the CRA can either underinvest or overinvest in information from a surplus perspective. If the CRA underinvests in accuracy, policies that increase market informativeness, such as improving disclosure, governance, or trust in credit-risk trading venues, can raise surplus by improving the discipline of the CRA. If the CRA overinvests, the same policies can reduce surplus because the extra information production is socially costly and can push rating accuracy further beyond the surplus-maximizing level. The paper therefore gives a market-specific view of regulation rather than a one-size-fits-all prescription.

2 Big picture

- **Research question.** Can prices in credit-risk markets discipline CRAs and improve rating accuracy?
- **Economic setting.** A CRA rates a risky project, investors decide whether to finance it, and a CDS-like market then generates a price signal about the project's credit risk.
- **Key friction.** The CRA is opportunistic with some probability and chooses how much costly information to acquire before issuing a rating.
- **Market signal.** A speculator may acquire costly information and trade against noise traders in the CDS market. The market maker sets a price from order flow.
- **CRA discipline.** More informative trading makes rating errors easier for investors to detect and therefore raises the CRA's reputational cost of inaccuracy.
- **Speculator incentive.** More accurate ratings reduce mispricing, so the speculator's incentive to acquire information falls when expected rating accuracy rises.
- **Equilibrium.** Rating accuracy and market informativeness are jointly determined by an upward-sloping CRA best response and a downward-sloping speculator best response.

- **Policy message.** Market discipline can supplement regulation, but direct regulation and market discipline interact; improving one may weaken the other.

3 Incremental contribution of the paper

- **Market discipline of rating agencies.** The paper gives a formal channel through which secondary credit-risk markets discipline CRAs, not only through investor trading payoffs but also through reputation.
- **Endogenous price informativeness.** The paper treats market informativeness as endogenous. It depends on the speculator's information acquisition, which itself depends on expected rating accuracy.
- **Two-sided feedback.** The same market signal has opposite incentive effects for the CRA and the speculator: it crowds in CRA effort but is crowded out by CRA effort.
- **Welfare ambiguity.** More information is not mechanically good. The CRA can underproduce or overproduce information relative to the surplus-maximizing level.
- **Policy interaction.** Reputational regulation can raise CRA accuracy but reduce market discipline; policies that foster informative trading can raise CRA accuracy indirectly.
- **Robustness.** The mechanism survives linked primary and secondary markets, multiple CRAs, moral hazard through rating inflation, and other sources of discipline such as project outcomes and media.
- **Credit-market specificity.** The market that disciplines the CRA must be informative about credit risk and rating accuracy. The discipline does not arise from any price signal in an arbitrary market.

4 Model setup

4.1 The ratings process

The issuer has a risky investment project. The project quality is

$$\theta \in \{B, G\},$$

where B denotes a bad project and G denotes a good project. The project succeeds or fails, with bad projects failing more often than good projects. Let the failure probabilities be

$$f_B \in (0, 1), \quad f_G \in (0, f_B), \quad \Delta f = f_B - f_G.$$

The project returns 1 in success and 0 in failure, and costs I to implement. Good projects have positive NPV, while the ex ante average project is negative NPV. Thus, without information production by a CRA, the project is not financed.

The CRA observes a private signal and offers a credit report

$$\tilde{r} \in \{H, L\},$$

where H is a high rating and L is a low rating. The issuer may purchase and publicize the report or decline to purchase it. If the report is not purchased, investors observe that the issuer is unrated. This creates rating shopping in the sense that the issuer can selectively publicize favorable ratings.

Investors observe the public rating outcome $r \in \{\tilde{r}, \emptyset\}$ and decide whether to buy the claim. The market price p^r equals investors' expected payoff from the project conditional on the rating. The project is implemented only if $p^r > I$.

4.2 The credit-risk market

If the project is implemented, a market for credit risk opens at time $t = 2$. The paper describes this as a CDS market, though the same logic applies to a secondary market for the asset. A CDS buyer pays price p^{cds} and receives one unit if the project defaults.

The CDS market has three types of participants:

- a noise or liquidity trader who trades $x_n \in \{-n, +n\}$;
- a second trader who is a speculator with probability η and another noise trader with probability $1 - \eta$;
- a competitive market maker who observes total order flow and sets the CDS price.

The speculator observes the rating and chooses the precision of her signal,

$$\iota \in \left[\frac{1}{2}, 1\right],$$

where

$$\iota = \Pr(\sigma_s = g \mid \theta = G) = \Pr(\sigma_s = b \mid \theta = B).$$

The cost of precision is $K(\iota) = \rho_s k(\iota)$. The speculator's information is valuable because it allows her to trade against noise in the CDS market.

4.3 CRA type and reputation

The CRA can be informative or opportunistic. The informative CRA observes the true project quality. The opportunistic CRA chooses how much information to acquire. Its rating accuracy is

$$\alpha \in \left[\frac{1}{2}, 1\right],$$

where α is the probability of observing the correct signal. The cost of accuracy is $C(\alpha) = \rho_{cra} c(\alpha)$.

The CRA receives a rating fee when the issuer buys the rating, and also receives a reputational payoff. Investors update their beliefs about the CRA's type after observing the public rating, CDS market information, and project outcome. This reputation channel is what gives the opportunistic CRA incentives to acquire information.

5 Timing of the model

1. $t = 0$: **quality and CRA type.** Nature chooses project quality $\theta \in \{B, G\}$ and CRA type $\tau \in \{I, O\}$. The CRA privately observes its type.

2. **$t = 1$: rating and initial investment market.** The CRA offers a report. The issuer decides whether to buy and publicize it. Investors decide whether to finance the project.
3. **$t = 2$: CDS market.** If the project is implemented, noise trading and speculative trading take place in the CDS market, and the market maker sets a CDS price from order flow.
4. **$t = 3$: project outcome and reputation.** The project outcome is realized. Investors observe the outcome and the CDS market signal and update beliefs about the CRA.

The equilibrium concept is Perfect Bayesian equilibrium. The model is solved backward: first the CDS market equilibrium, then the CRA's rating accuracy choice, and finally the joint equilibrium.

6 The CDS market equilibrium

6.1 Trading strategy and price signal

Given a high rating and a conjecture about CRA accuracy, the speculator trades on her signal. If she observes bad news about the project, she buys CDS protection; if she observes good news, she sells protection. The market maker observes total order flow, not trader identities, and sets the CDS price using Bayes' rule.

The important equilibrium object is the level of market informativeness,

$$\iota^*(\alpha),$$

which is the speculator's chosen precision as a function of expected rating accuracy α .

6.2 Strategic substitute from the speculator's perspective

Lemma 3 shows that the CDS market equilibrium is unique. It also shows that $\iota^*(\alpha)$ decreases with expected rating accuracy. This is the first core strategic interaction.

The intuition is the mispricing channel. If the rating is expected to be less accurate, CDS prices are more likely to be wrong. A speculator can profit more from learning the true project quality. If the rating is already very accurate, there is less mispricing to exploit, so information acquisition in the CDS market is less attractive.

Thus, from the speculator's perspective,

$$\text{rating accuracy } \uparrow \Rightarrow \text{market informativeness } \downarrow.$$

7 The rating game

7.1 CRA payoff

The opportunistic CRA chooses rating accuracy before offering a rating. Its payoff combines the rating fee, reputation, and the cost of information. A simplified version of the incentive problem is:

$$\Pi^{cra} = \text{expected rating fee} + \gamma \cdot \text{expected reputation} - C(\alpha).$$

The parameter γ measures the weight the CRA places on reputation.

Higher accuracy helps the CRA because it increases the probability that a good project receives a high rating and a bad project does not. More importantly for the paper's mechanism, higher accuracy improves the CRA's future reputation when the CDS price and project outcome reveal that the rating was correct.

7.2 Strategic complement from the CRA's perspective

Lemma 4 shows that, for a given conjecture about market informativeness ι , the rating game is unique and the equilibrium rating accuracy $\alpha^*(\iota)$ increases with expected market informativeness.

The intuition is transparency. When the CDS market is more informative, rating errors become easier to detect. If an opportunistic CRA issues an inaccurate rating, the CDS order flow and price are more likely to reveal the mistake. Because investors use these signals to update beliefs about the CRA's type, a more informative market strengthens the reputational penalty for inaccuracy.

Thus, from the CRA's perspective,

$$\text{market informativeness } \uparrow \Rightarrow \text{rating accuracy } \uparrow.$$

This is the crowding-in effect: informative trading crowds in CRA accuracy.

8 Joint equilibrium

The joint equilibrium combines the two best-response functions:

$$\alpha = \alpha^*(\iota), \quad \iota = \iota^*(\alpha).$$

The CRA's best response is upward sloping: more market informativeness raises rating accuracy. The speculator's best response is downward sloping: higher expected rating accuracy reduces market informativeness.

Proposition 1 establishes the unique equilibrium pair (α^*, ι^*) . The equilibrium exists with an active rating and credit-risk market only if rating accuracy is high enough to make a high-rated project financeable. If equilibrium accuracy is too low, the project is not implemented, and both the rating market and the CDS market fail to exist.

Figure 1 gives the graphical representation. The downward-sloping curve is the speculator's best response $\iota^*(\alpha)$, and the upward-sloping curve is the CRA's best response $\alpha^*(\iota)$. Their intersection gives the unique equilibrium.

9 Comparative statics

Lemma 5 gives the main comparative statics.

First, increasing the volume of noise trading n raises market informativeness because the speculator can better camouflage informed trades and earn more profit. This increases ι^* and, through the crowding-in effect, raises α^* as well. A decrease in the speculator's information cost ρ_s has the same effect.

Second, increasing the weight on reputation γ raises rating accuracy directly. But because higher rating accuracy reduces mispricing, it decreases market informativeness. A decrease in the CRA's information cost ρ_{cra} has the same effect.

Third, a common increase in information costs can reduce rating accuracy and have ambiguous effects on market informativeness.

These comparative statics are economically useful. Low trading volume, market fragmentation, and crises can reduce CDS market informativeness. The CRA does not fully substitute for the lost market discipline. Instead, lower market informativeness weakens CRA incentives and can lower rating accuracy exactly when accurate ratings are especially valuable.

10 Real effects and welfare

The paper then adds real effects of information. Rating accuracy has direct real effects because it affects whether projects are financed. CDS market information has indirect real effects because it disciplines CRA accuracy. The surplus function includes the NPV of projects that receive high ratings and the information costs borne by the CRA and speculator.

Proposition 2 shows that the equilibrium CRA accuracy can be below or above the surplus-maximizing level. This is why the policy implications are nuanced.

If the CRA underinvests in information, increasing reputational discipline or market discipline can raise surplus. In that case, policies that reduce speculator information costs or increase the depth of the credit-risk market improve welfare by pushing α^* closer to the surplus-maximizing level.

If the CRA already overinvests in information, more market discipline can reduce surplus. The cost of extra information production can dominate the benefit of more accurate screening. Hence, the same policy can help in one market and hurt in another.

11 Extensions

11.1 Linked primary and secondary markets

The baseline separates the initial investment market from the secondary credit-risk market. Section 6.1 links them by allowing investors in the primary market to trade in the CDS market later because of liquidity shocks. Proposition 3 shows that the equilibrium continues to exist and remains unique. The trading and pricing logic is the same as in the baseline.

The new feature is that liquidity costs in the secondary market affect willingness to pay in the primary market. Investors know they may later need to trade against an informed speculator. This reduces the price they are willing to pay initially and makes implementation harder. Therefore, connecting the primary and secondary markets preserves market discipline but raises the threshold for the rating and CDS markets to exist.

11.2 Multiple rating agencies

Section 6.2 introduces two CRAs. Lemma 8 shows that the issuer publicizes ratings and undertakes the project only when both ratings are high. Split ratings are not publicized because one high and

one low rating leaves investors' valuation too low. This helps explain why rating disagreement may be rarely observed in rated issues.

Proposition 4 shows that the CDS market still disciplines CRAs in the multiple-CRA setting. However, competition among CRAs is not a simple substitute for market discipline. The CRAs' accuracy choices can be strategic complements or substitutes, and equilibrium accuracy can be higher or lower than in the single-CRA baseline.

The important economic point is that more CRAs do not necessarily solve the accuracy problem. If the issuer only buys and publicizes ratings when all ratings are high, investors may learn less about each CRA's type. This can weaken reputational incentives.

11.3 Inflated ratings

Section 6.3 allows the opportunistic CRA to inflate ratings. Instead of always reporting its signal truthfully, the CRA may stay uninformed and offer a high rating. Proposition 5 characterizes mixed-strategy equilibria in which the CRA may randomize between acquiring information and truthfully revealing it, or remaining uninformed and offering a high rating.

The CDS market still disciplines the CRA. When market informativeness rises, inaccurate or inflated high ratings are more likely to be exposed. This makes pure rating inflation less attractive and makes equilibria with information acquisition more likely.

Lemma 9 further shows that market informativeness enlarges the parameter region in which the CRA acquires information. A high enough reputational payoff, or a low enough rating fee from inflated ratings, can fully discipline the CRA so that it always acquires information.

11.4 Project outcomes, other CRAs, and media

Section 7 compares market discipline to other sources of discipline.

Project outcomes eventually reveal information about project quality. Lemma 10 shows that increasing the information sensitivity of outcomes, Δf , always increases rating accuracy, but can raise or lower market informativeness. Project outcomes are useful, but defaults may be observed only after long delays, while the CDS market provides continuous and interim information.

The paper also compares the disciplining role of an extra CRA to that of the CDS market. Proposition 6 shows that two CRAs without a credit-risk market can generate more or less surplus than one CRA with credit-risk market information. Thus, competition among CRAs and market discipline are different governance mechanisms.

Media can also reveal information, but the paper emphasizes that credit-risk market information is distinctive because it is directly tied to the rated asset's credit quality and is generated through trading incentives.

12 Empirical implications

- **Credit-risk market depth.** Ratings should be more accurate when the rated issuer or asset has an active and liquid CDS or secondary bond market.

- **Market fragmentation.** Ratings should be less accurate when credit-risk trading is fragmented, inactive, or costly for informed traders.
- **Noise trading and liquidity.** More uninformed trading can increase informed trading profits and thereby improve price informativeness, which can discipline the CRA.
- **Regulation and market discipline.** Stronger direct CRA regulation may increase accuracy directly but reduce the information produced in the CDS market.
- **Multiple CRAs.** The presence of several CRAs is not necessarily enough to discipline rating accuracy, especially when issuers selectively publicize favorable ratings.
- **Rating inflation.** More informative credit-risk prices should reduce incentives to issue inflated high ratings.
- **Defaults as discipline.** Project outcomes discipline ratings, but if defaults occur with delay, market prices can provide faster reputational feedback.

13 How to use the paper

Use this paper when studying information intermediaries whose incentives depend on downstream price signals. The cleanest application is to CRAs, CDS markets, and secondary bond markets, but the broader logic applies whenever an intermediary's report is followed by a market that can reveal whether the report was accurate.

The paper is also useful for thinking about regulation. It shows why a regulator should not evaluate CRA rules in isolation. A policy that directly increases CRA accuracy may reduce informed trading and weaken market-based monitoring. Conversely, policies that improve trading transparency or market liquidity may improve rating accuracy even though they do not directly regulate CRAs.

For a finance research discussion, the most important phrase is **endogenous market discipline**. The market price disciplines the rating agency, but the agency's own accuracy changes the informativeness of the market price.

14 Limitations and caveats

- **The paper is theoretical.** The key outputs are mechanisms, comparative statics, and empirical implications rather than direct causal estimates.
- **CDS market structure is stylized.** The market is modeled in a simplified Kyle-style setting with noise traders, a speculator, and a market maker.
- **Reputation is reduced-form.** The model captures reputational incentives through posterior beliefs and payoff weights, not through a fully specified dynamic CRA market.
- **Rating shopping is simplified.** The issuer's selective purchase/publication decision captures an important feature of ratings, but real rating contracts and regulatory disclosure rules can be more complex.
- **Policy effects are parameter-dependent.** The same change in market informativeness can raise or lower surplus depending on whether the CRA underinvests or overinvests in accuracy.

- **External validity depends on market observability.** The discipline mechanism is strongest when market prices are observable, informative, and clearly linked to the rating's underlying credit risk.

15 Original paper extracts

15.1 Title, model environment, and timing

Read: The first pair shows the title/abstract and the model's rating process. The second pair shows the CDS market and the timing of the game.

Economic message: The paper explicitly connects a ratings market to a later credit-risk market. That sequence is what allows CDS prices to discipline rating accuracy.

Credit Ratings and Market Information

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Accurate credit ratings are important for both investors and regulators. We demonstrate that the market for credit risk provides an important source of discipline for credit rating agencies (CRAs). We examine a model in which a CRA's rating is followed by a market for credit risk that provides a public signal – the price. More informative trading increases the CRA's incentives to be accurate by making rating errors more transparent. We show that this source of discipline is (a) robust to moral hazard, multiple CRAs, and connected primary and secondary markets and (b) specific to the market for credit risk. (*JEL* G24, G28)

(a) Title and abstract

1. The Model

Our model has two distinct elements, the ratings process and the market for credit risk. We first present them separately and then analyze the strategic interactions between them. The market for ratings takes place first, and is followed by the market for credit risk. All agents in the model are rational and risk neutral, and, for simplicity, we assume that there is no discounting.

1.1 The ratings process

The ratings process takes place at time $t=1$, and consists of three types of agents: an issuer, a monopoly credit rating agency (CRA), and investors.

The issuer has access to a risky investment project but needs financing from investors. The project returns 1 in case of success and 0 in case of failure, and the cost of the project is I . $y \in \{S, F, N\}$ denotes the project's outcome, where S (F) signifies *Success* (*Failure*) and N represents the case in which the project is not undertaken. The quality of the project is denoted by $\theta \in \{B, G\}$, where B (G) stands for *Bad* (*Good*) and relates to its probability of failure: a bad project fails with probability $f_B \in (0, 1)$, and a good project fails with probability $f_G \in (0, f_B)$, with $f_B - f_G \equiv \Delta^f$. Both good and bad investments have an ex ante probability $\frac{1}{2}$ of occurring, and the investment quality is *a priori* unknown, even to the issuer itself.

If the issuer wishes to undertake the project, it can sell a claim to the payoff of the project to investors. $v_\theta = 1 - f_\theta - I$ denotes the net present value (NPV) of a project of quality θ . Good projects should be financed (they have positive NPV, i.e., $v_G > 0$) but, without prior knowledge about the quality of the project, no financing takes place (ex ante NPV is negative, i.e., $\frac{1}{2}v_G + \frac{1}{2}v_B < 0$). Therefore, the presence of a CRA can improve welfare by screening projects for investors.

The CRA observes a signal of project quality (more on this shortly) and privately offers a credit report, $\tilde{r} \in \{H, L\}$, where H signifies *high* and L signifies *low*, to the issuer. The issuer either pays a rating fee φ and has the report publicized or refuses to purchase it. This allows for “rating shopping” by the issuer. The outcome of the rating process, as observed by investors, is thus $r \in \{\tilde{r}, \emptyset\}$, where $r = \tilde{r}$ signifies that the issuer had the credit report publicized by the CRA, and $r = \emptyset$ signifies that there is no rating. If the issuer refuses to buy the CRA's report and goes on the market as unrated, that in itself is a signal to investors.

Investors observe the rating and decide whether they wish to buy the claim. p^r denotes the price investors are willing to pay for a claim backed by the payoff of the project for a given rating r . Investors are perfectly competitive, so

(b) Rating process and project setup

Appendix B.9, we show that these results continue to hold in a setting where the rating fee is endogenous.

1.2 The market price of risk

At time $t=2$, if the issuer implemented the investment project, a market for credit risk takes place. We will describe this market as the CDS market in what follows. The same setting can be used to model the secondary market for the asset.⁸

p^{cds} denotes the price of a CDS contract and x the net volume of trades.⁹ A CDS contract is formalized as follows: at time $t=2$, the contract is signed, and the buyer of the swap pays an amount p^{cds} to the swap's seller. In return, the seller agrees that in the event of default at time $t=3$, the seller will pay the buyer an amount 1.¹⁰ Trading occurs among a noise/liquidity trader, who trades an amount $x_n \in \{-n; +n\}$, with both realizations equally likely; a second trader, who may be a speculator or another noise trader (more on this shortly); and a competitive market maker. The price p^{cds} is determined in a simplified model à la Kyle (1985).

With probability $\eta \in (0, 1]$, the second trader is a speculator, who acquires information to profit off the noise trader in the market. Having observed the rating r , a speculator decides how much information about the project to acquire. She chooses the precision $\iota \in [\frac{1}{2}, 1]$ of her private signal $\sigma_s \in \{b, g\}$, where

$$\iota \equiv \Pr(\sigma_s = g | \theta = G) = \Pr(\sigma_s = b | \theta = B).$$

$K(\iota) = \rho_s k(\iota)$ denotes the cost of precision, and we assume $\rho_s > 0; k' \geq 0; k'' > 0$; and $k(\frac{1}{2}) = k'(\frac{1}{2}) = 0$. To ensure that the equilibrium choice of ι is unique, we

⁸ The easiest way to model this as a secondary market is to endow the speculator with some of the asset. Allowing for shorting would be equally good, of course.

⁹ In a CDS contract, a protection buyer pays a premium to the protection seller in exchange for a payment from

The timing of the model is summarized below:

Time $t=0$ (Quality of the investment and CRA type): Nature chooses the quality of the investment project $\theta \in \{B, G\}$ and the CRA's type $\tau \in \{I, O\}$. The CRA privately observes τ .

Time $t=1$ (Rating and initial investment market):

- The CRA offers a report $\tilde{r}$ to the issuer.
- The issuer either agrees to pay the rating fee and has the report publicized ($r = \tilde{r}$) or refuses to purchase it and goes on the market as unrated ($r = \emptyset$).
- Given the rating r , the issuer sets a price p^r and markets a claim backed by the investment project to investors.
- If $p^r > I$, the issuer sells the claim and undertakes the project. If the report was publicized, the issuer pays the CRA.

Time $t=2$ (CDS market): If the issuer undertakes the project, a noise trader, a second trader, and a market maker trade CDS contracts on the outcome of the project.

Time $t=3$ (Investment outcome and reputational payoff): The investment outcome y is realized. Having observed (r, x, y) , investors update their beliefs about the CRA's type.

We use the Perfect Bayesian equilibrium as the solution concept.

(a) Market for credit risk

(b) Timing of the model

15.2 Strategic interaction between the CRA and the CDS market

Read: Lemma 3 gives the speculator's response: market informativeness falls when expected rating accuracy rises. Lemma 4 and the following discussion give the CRA response: rating accuracy rises when market informativeness rises.

Economic message: The speculator and CRA face opposite incentives. Accurate ratings reduce trading opportunities, while informative trading raises reputational discipline.

1. The equilibrium trading strategies are as described in lemmas 1 and 2.
2. The equilibrium level of precision $\iota^*(\bar{\alpha})$ is unique, with $\iota^*(\bar{\alpha}) \in (\frac{1}{2}, 1)$ if $\bar{\alpha} < 1$ and $\iota^*(1) = \frac{1}{2}$.
3. The equilibrium level of precision $\iota^*(\bar{\alpha})$ decreases with expected rating accuracy $\bar{\alpha}$.

Proof: See Appendix A.3.

Interestingly, from the speculator's point of view, information acquisition and rating accuracy are strategic substitutes. That is, lower expected rating accuracy increases the incentive to acquire information. This occurs through the mispricing channel, as the speculator can take advantage of wrong valuations due to less precise ratings.²²

3. The Rating Game

Having characterized the equilibrium trading strategies and level of precision in the CDS market, we can now characterize the equilibrium rating accuracy for a given level of expected precision $\bar{\iota}$. The properties of the strategic interaction

between credit ratings and the market for credit risk will be used in the next section to characterize the unique equilibrium of the game.

At time $t=1$, before offering a rating to the issuer, an opportunistic CRA decides how much information to acquire about the project. We can write the CRA's total payoffs as follows:

$$\begin{aligned} \Pi^{CRA} = & \frac{1}{2} \{ \alpha [\varphi + \gamma E[q_{H,x,y} | \theta = G]] + (1 - \alpha) \gamma q_{\emptyset} \} \\ & + \frac{1}{2} \{ (1 - \alpha) [\varphi + \gamma E[q_{H,x,y} | \theta = B]] + \alpha \gamma q_{\emptyset} \} - C(\alpha). \end{aligned} \quad (4)$$

The ex ante probability that the project is good, that is, $\theta = G$, is $\frac{1}{2}$. In this case, with probability α , the CRA observes a positive signal $\sigma_{CRA} = g$ and offers a high

²² The speculator benefits more from observing a correct (negative) signal and trading on it when $\theta = B$ as the

(a) Lemma 3: CDS market equilibrium

(b) Rating game payoff

1. In equilibrium, an opportunistic CRA chooses a level of rating accuracy $\alpha^*(\bar{t}) \in (\frac{1}{2}, 1)$.
2. The equilibrium level of rating accuracy α^* increases with expected market informativeness $\bar{t}$.

Proof: See Appendix A.4.

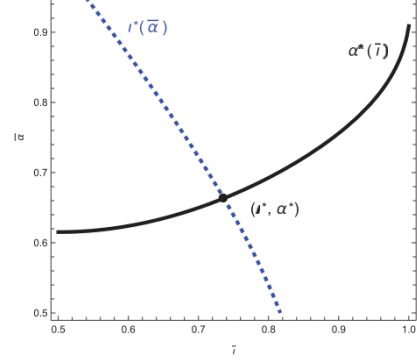
This indicates that from the point of view of the CRA, rating accuracy and market trading informativeness are strategic complements. That is, more informative market trading incentivizes the CRA to be more accurate. This arises because market transparency from informative trading makes inaccurate ratings more transparent, thus making reputational incentives more important.

We now examine this effect in detail. The effect of market informativeness on $\alpha^*(\bar{t})$ depends on how $\frac{\partial \Pi^{CRA}}{\partial \alpha}$ changes with $\bar{t}$. Taking the derivative of $\frac{\partial \Pi^{CRA}}{\partial \alpha}$ with respect to $\bar{t}$ yields

$$\frac{\partial \Pi^{CRA}}{\partial \alpha \partial \bar{t}} = \frac{\gamma}{2} \left\{ \frac{\partial E[q_{H,x,y} | \theta = G]}{\partial \bar{t}} - \frac{\partial E[q_{H,x,y} | \theta = B]}{\partial \bar{t}} \right\}. \quad (6)$$

When $\bar{t}$ increases, the distribution of x is expected to be more informative about the true project quality, since the speculator is more likely to observe the correct signal and sell CDS protection when $\theta = G$ and buy it when $\theta = B$. This implies that, conditional on $\theta = G$ ($\theta = B$), positive news about project quality becomes more (less) likely. As a result, $E[q_{H,x,y} | \theta = G]$ increases and

(a) CRA accuracy and market informativeness



rating accuracy and trading informativeness

e, $C(\alpha) = (\alpha - \frac{1}{2})^2$, $K(t) = (t - \frac{1}{2})^2$, $q_0 = \frac{1}{2}$, $\gamma = 3$, $\eta = n = 1$, $f_B = 0.6$, $f_G = 0.2$.

(b) Figure 1: equilibrium best responses

15.3 Equilibrium and comparative statics

Read: Proposition 1 gives the unique equilibrium. Lemma 5 gives the comparative statics for noise trading, information costs, and reputational concerns.

Economic message: More informative credit-risk trading can indirectly improve ratings. But more direct CRA discipline can crowd out market informativeness.

in this example, $C(\alpha) = (\alpha - \frac{1}{2})^2$, $K(t) = (t - \frac{1}{2})^2$, $q_0 = \frac{1}{2}$, $\gamma = 3$, $\eta = n = 1$, $f_B = 0.6$, $f_G = 0.2$.

follows that the equilibrium of the game is unique, and is characterized as follows:

1. If the equilibrium level of rating accuracy is larger than a threshold, that is, if

$$\alpha^* > \underline{\alpha} \equiv \left(\frac{-v_B}{v_G - v_B} - q_0 \right) (1 - q_0)^{-1},$$

then the investment project is implemented following a high rating ($p^H > I$), and both the opportunistic CRA and the speculator produce information about the project's quality, that is, $\alpha^* \in (\max\{\underline{\alpha}, \frac{1}{2}\}, 1)$ and $t^* \in (\frac{1}{2}, 1)$.

2. Otherwise, the project is not implemented, and the market for ratings and the CDS market fail to exist.

Proof: See Appendix A.5.

(a) Proposition 1: unique equilibrium

find:

Lemma 5. The following results hold in equilibrium:

1. An increase in the volume of the noise traders' demand, n , increases both equilibrium market informativeness, t^* , and equilibrium rating accuracy, α^* . The same results apply to a decrease in the speculator's cost of information (ρ_s).
2. An increase in the weight on reputational payoffs in the CRA's objective function (γ) decreases t^* and increases α^* . The same results apply to a decrease in the CRA's cost of information (ρ_{CRA}).
3. Assume $\rho_{CRA} = \rho_s = \rho$; an increase in ρ decreases α^* , while it may increase or decrease t^* .

Proof: See Appendix A.6.

The comparative statics on the volume of noise trading n are straightforward. When n increases, the speculator can trade more and, thus, makes more profits. Therefore, she chooses a larger $t^*(\bar{\alpha})$ for any level of $\bar{\alpha}$ ($t^*(\bar{\alpha})$ rotates counterclockwise).²⁴ Note that the volume of noise trading does not enter the CRA's objective function, so that a change in n affects the equilibrium only through its direct effect on $t^*(\bar{\alpha})$. This implies that the new equilibrium pair will feature a higher level of precision in the CDS market and higher rating accuracy. Similar results apply to a decrease in the cost of precision for the speculator.

This comparative static has an intriguing implication. When the volume of trading is low (e.g., during a crisis or when markets are fragmented), the

(b) Lemma 5: comparative statics

15.4 Real effects and linked markets

Read: Proposition 2 summarizes the welfare logic. Proposition 3 shows that the mechanism survives when the primary and secondary markets are connected.

Economic message: Market discipline is valuable when rating accuracy is too low, but more market informativeness is not always welfare-improving.

increase surplus because of the discipline it provides to the CRA's incentives.

Proposition 2. $\hat{\alpha}$ denotes the level of rating accuracy that maximizes the expected total surplus S . The following comparative statics results hold in equilibrium:

1. Equilibrium rating accuracy is smaller than $\hat{\alpha}$ at time $t=3$ (i.e., $\alpha_3^* = \frac{1}{2} < \hat{\alpha}$), while it may be larger or smaller than $\hat{\alpha}$ at time $t=1$.
2. If equilibrium rating accuracy is lower than $\hat{\alpha}$ at time $t=1$ (i.e., if $\alpha_1^* < \hat{\alpha}$), we have:
 - (a) S is increasing in the weight on reputational payoffs in the CRA's objective function (γ) and decreasing in the CRA's cost of information (ρ_{cra}).
 - (b) If the speculator's cost of information ρ_s is sufficiently small, S is decreasing in ρ_s and increasing in the volume of the noise trader's demand (n).
3. If $\alpha_1^* \geq \hat{\alpha}$, S is decreasing in n , and may increase or decrease with γ , ρ_{cra} , and ρ_s .

Proof: See Appendix A.8.

The equilibrium rating accuracy at time $t=1$ (α_1^*) is pinned down by Equation (8). The left-hand side of this equation describes the CRA's private benefit of accuracy; α_1^* is lower than its optimal level $\hat{\alpha}$, if the CRA's weight on reputational payoffs γ or market informativeness ι^* are small. In this case, increasing α_1^* directly, via an increase in γ , increases surplus. Perhaps more surprisingly, if the marginal cost of ι is sufficiently small, increasing α_1^* indirectly, via an increase in market discipline ι^* , also increases surplus. In this case, the marginal cost of an increase in ι^* is lower than the marginal

lemma describes the equilibrium in this extension of the game.

Proposition 3. An equilibrium of the game where the primary and secondary markets are connected always exists, and is unique. In this equilibrium, we have:

1. Investors with a liquidity shock at time $t=2$ ($t=3$) sell (acquire) a unit of CDS protection; their aggregate demand is $x_n = -n$ if $z=(\ell, 0)$ and $x_n = +n$ if $z=(0, \ell)$.
2. The speculator's trading strategy and the market maker's pricing function are the same as in lemmas 1 and 2, respectively.
3. The equilibrium pair of rating accuracy and market trading informativeness ($\alpha^{link}, \iota^{link}$) is characterized as in Proposition 1, except that the market for ratings and the CDS market fail to exist if $\alpha^{link} \leq \underline{\alpha}^{link}$, where $\underline{\alpha}^{link} > \underline{\alpha}$.

Proof: See Internet Appendix B.1.

In equilibrium, all investors acquire a share of the claim if the project is implemented. As a consequence, the aggregate demand of CDS protection from the investors who face a liquidity shock (*liquidity traders*) is $x_n \in \{-n, +n\}$, with both realizations equally likely (according to the distribution of z). Since the distribution of x_n is the same as in the main model, the speculator's and

(a) Proposition 2: surplus and real effects

(b) Proposition 3: linked markets

15.5 Multiple CRAs and rating inflation

Read: Lemma 8 explains disclosure with two CRAs. Proposition 4 shows that CDS discipline remains in the multiple-CRA case. Proposition 5 shows how the model handles inflated ratings.

Economic message: Competition among rating agencies is not a complete substitute for price-based discipline. The issuer's selective disclosure creates strategic effects among CRAs.

can seek ratings from two different CRAs ($j \in \{A, B\}$). Since the modeling of the CDS market does not change, we now elaborate on the interaction between the issuer and the CRAs.

We make the following assumptions.³⁴ The issuer can seek ratings from both CRAs. Each of the two CRAs is identical to the CRA in the baseline model. The realizations of the CRAs' types and of their signals are independent across CRAs. α_j denotes an opportunistic CRA j 's level of rating accuracy. Each CRA j observes a signal about project quality and privately offers a rating $\tilde{r}_j \in \{H, L\}$ to the issuer. The issuer either pays a fee to the CRA and has the rating publicized or refuses to purchase it. If a rating is publicized, it becomes public knowledge. The issuer can observe both offers $\tilde{r}_A$ and $\tilde{r}_B$ at no cost and then decide which ratings to publicize. Investors only observe the publicized ratings $r = (r_A, r_B)$, namely, one of four possible cases: a project with two ratings; a project only rated by A ; a project only rated by B ; or an unrated project. Therefore, we have $r \in \{(\tilde{r}_A, \tilde{r}_B), (\tilde{r}_A, \emptyset), (\emptyset, \tilde{r}_B), (\emptyset, \emptyset)\}$, where $\emptyset$ signifies the project is *unrated*. Since the CRAs are ex ante identical, here we restrict our attention to symmetric equilibria, that is, equilibria in which the two CRAs choose the same level of rating accuracy. In Internet Appendix B.10, we show that our main qualitative results continue to hold in the case of asymmetric equilibria. Like in the main model, we assume that the rating fee is a fraction of the issuer's surplus from implementing the project: $\varphi_A = \varphi_B = \frac{\phi}{2} (p^{\tilde{r}_A, \tilde{r}_B} - I)$ if the issuer buys ratings from both CRAs; and $\varphi_j = \phi (p^{\tilde{r}_j, \emptyset} - I)$ if the issuer only buys the rating from CRA j .

The following lemma describes the issuer's disclosure strategy and the rating fees.

Lemma 8. In a symmetric equilibrium of the rating game with two CRAs, the issuer publicizes the ratings, sells the claim, and undertakes the project only when both ratings are high ($r = (H, H)$); otherwise, the claim is unrated ($r = (\emptyset, \emptyset)$) and the project is not undertaken. Proof: See Internet Appendix B.3.

Since the NPV of the asset is ex ante negative, the issuer needs two high ratings to sell the claim and implement the project. Split ratings (one high and one low) would cancel each other out and leave investors' valuation of the project negative. As a consequence, split ratings are never publicized by the issuer and, thus, never observed by investors. This result is consistent with stylized facts in the rating industry, where disagreement among CRAs is rarely

(a) Lemma 8: two CRAs and rating disclosure

to hold.

To make this extension tractable, we assume that the opportunistic CRA cannot deflate ratings; that is, it cannot offer a low rating to the issuer after having observed a good signal. Moreover, we restrict our attention to equilibria where the opportunistic CRA always offers a H rating when it chooses not to acquire information about the project.

The following proposition describes the equilibrium rating strategy. For presentational purposes, we let the CRA's cost of accuracy be such that $C'(1) > \gamma/2$ and simplify the main model slightly by assuming the CRA's reputational payoff is only conditional on the realization of market trading x , but not on the realization of the project outcome, y .^{38,39} The modeling of the CDS market does not change here, so we focus on the CRA's incentives in this section.

Proposition 5. For a given conjecture about market informativeness $\bar{\lambda}$, an equilibrium of the rating game where the CRA inflates its rating always exists, and there may be more than one. In equilibrium, an opportunistic CRA mixes between acquiring information (i.e., $\alpha^{infl} \in (\frac{1}{2}, 1)$) and truthfully revealing it to investors (with probability $\lambda^{infl} \in [0, 1]$), and staying uninformed and offering a high rating to the issuer (with probability $1 - \lambda^{infl}$), where α^{infl} is such that

$$\frac{\gamma}{2} \{E[q_{H,x} | \theta = G] - E[q_{H,x} | \theta = B]\} |_{\bar{\alpha} = \alpha^{infl}, \bar{\lambda} = \lambda^{infl}} = C'(\alpha^{infl}). \quad (11)$$

(a) Proposition 5: rating inflation, part 1

Proposition 4. For a given conjecture about market informativeness $\bar{\lambda}$, a symmetric equilibrium of the rating game with two CRAs always exists, and there may be more than one; moreover:

1. There always exists a symmetric equilibrium in which both opportunistic CRAs' level of rating accuracy weakly increases with the conjecture about the speculator's signal precision $\bar{\lambda}$.
2. The equilibrium accuracy α_j^{mult} may increase or decrease with j 's conjecture about an opportunistic CRA $-j$'s level of rating accuracy ($\bar{\alpha}_{-j}$), and may be larger or smaller than the equilibrium rating accuracy in the baseline model (α^*).

Proof: See Internet Appendix B.4.

Proposition 4 offers three main insights. First, the speculator continues to discipline the CRAs' accuracy incentives in this extension of the model.³⁶ Second, the CRAs' investments in rating accuracy may be strategic substitutes or strategic complements; we discuss the intuition below. Third, the implication of the strategic interaction between the CRAs is that their incentives to invest in accuracy may be weakened, leading to lower accuracy choices than in a setting with a single CRA. This draws a contrast between the role of the CDS market and that of competition in disciplining a CRA's incentives to be accurate. Section 7.2 explores this contrast further by comparing the surplus in both cases.

We now examine the strategic interaction between the two CRAs further. Since the issuer only publicizes two high ratings, both CRAs must offer a high rating for each one to sell its rating. The more accurate the ratings, the higher the chance they match. This relation can be explained by the fact that precise

(b) Proposition 4: multiple CRAs

tunistic CRA is informed with strictly positive probability in equilibrium (i.e., $\lambda^{infl} > 0$). Otherwise, the CRA always stays uninformed and offers a high rating to the issuer (i.e., $\lambda^{infl} = 0$).

2. If $\{\gamma q_{\emptyset} - (\varphi^{infl} + \gamma E[q_{H,x}])\} |_{\bar{\alpha} = \alpha^{infl}, \bar{\lambda} = 1} > 0$ (**Condition 2**), there exists an equilibrium where the CRA is always informed (i.e., $\lambda^{infl} = 1$).
3. If Condition 1 holds but Condition 2 does not hold, the equilibrium features $\lambda^{infl} \in (0, 1)$, where λ^{infl} is such that

$$\{\gamma q_{\emptyset} - (\varphi^{infl} + \gamma E[q_{H,x}])\} |_{\bar{\alpha} = \alpha^{infl}, \bar{\lambda} = \lambda^{infl}} = 0. \quad (12)$$

The expressions for $q_{H,x}$, $q_{\emptyset}$, and φ^{infl} are described in the Internet Appendix. If α^{infl} and λ^{infl} are such that $p^H > I$, the investment project is implemented following a high rating. Otherwise, the project is not implemented and the market for ratings and the CDS market fail to exist. Proof: See Internet Appendix B.5.

(b) Proposition 5: rating inflation, part 2

15.6 Other sources of discipline and appendix figures

Read: Lemma 10 and Proposition 6 compare market discipline to project outcomes and multiple CRAs. Figures A1 and A2 show the two best-response relations behind Figure 1.

Economic message: Project outcomes and CRA competition can discipline ratings, but they do

not remove the role of credit-risk market prices.

We have shown that the information produced by the speculator in the CDS market disciplines the CRA by providing investors with an informative signal about asset quality that they can use to assess the CRA's performance in rating the asset. In this section, we describe some other signals of asset quality that might be available to investors, and discuss the differences and similarities in their effects on the CRA's incentives.

7.1 Project outcomes

In the model, investors learn about the quality of the rating from two signals of asset quality: the volume of trades in the CDS market (x) and the realization of the project outcome (y). The informativeness of y depends on the relative likelihood of default across project qualities, which is measured by the asset's information sensitivity $\Delta^f = f_B - f_G$.

Lemma 10. An increase in the asset's information sensitivity Δ^f always increases equilibrium rating accuracy α^* , while it may increase or decrease equilibrium market informativeness $\bar{t}$. Proof: See Internet Appendix B.7.

When Δ^f goes up, y becomes a more informative signal of project quality, since bad assets fail relatively more often and good assets relatively less often. This makes inaccurate ratings ex post more transparent and, thus, disciplines the CRA. As a result, the increase in Δ^f has two contrasting effects on the speculator's incentives to acquire information. On the one hand, the value of the CDS contract is more sensitive to the quality of the underlying asset, which makes learning about the asset more profitable for the speculator. On the other hand, the increase in rating accuracy reduces mispricing and, thus, profits from informed trading. As a consequence, α^* always increases with Δ^f , while the effect of Δ^f on $\bar{t}$ is ambiguous.

In the limit as f_B approaches 1 and f_G approaches zero, a project's outcome becomes a perfect signal of its quality and, thus, investors only use y to assess the accuracy of the rating. However, there are two caveats to the disciplining role of y . First, defaults are unlikely to be observed in the short term, while trading in the CDS market is continuously providing information.^{41,42} Second, although y is eventually realized and observed in the long term, the CDS market

(a) Lemma 10: project outcomes

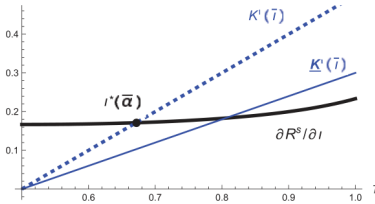


Figure A1
Equilibrium market informativeness for a given level of rating accuracy
As example, $K(t) = (t - \frac{1}{2})^2$, $q_0 = \frac{1}{2}$, $\bar{\alpha} = \frac{2}{3}$, $\eta = 1$, $n = \frac{3}{4}$, $f_B = 0.6$, $f_G = 0.2$.

can lead to a contradiction. Notice also that $\frac{\partial R^S}{\partial t}$ crosses $K'(t)$ at some interior level $\bar{t} \in (\frac{1}{2}, 1)$.

(a) Figure A1: market informativeness best response

in the main model, that is, $\alpha^{mult} \leq \alpha$. This difference in the optimal level occurs because two CRAs screen the projects here, and this screening improves the average quality of the projects that are implemented for a given rating accuracy.

The next proposition describes comparative statics results on S_{mult} .⁴⁶

Proposition 6. $\hat{\alpha}^{mult}$ denotes the level of rating accuracy that maximizes the expected total surplus, S_{mult} . The following comparative statics results hold in equilibrium:

1. Equilibrium rating accuracy at time $t = 1$ (α_1^{mult}) may be larger or smaller than $\hat{\alpha}^{mult}$. If $\alpha_1^{mult} < \hat{\alpha}^{mult}$ and the speculator's cost of information ρ_s is sufficiently small, S_{mult} is decreasing in ρ_s and increasing in the volume of the noise trader's demand (n).
2. Assume that the CRAs and the speculator have the same cost function, that is, $C(\alpha) = K(t)$ for any $\alpha = t$. Surplus when there are two CRAs and no information provided by the CDS market (i.e., S_{mult} evaluated at $\eta = 0$) may be larger or smaller than surplus in the model with one CRA with information provided by the CDS market (i.e., S evaluated at $\eta > 0$).

Proof: See Internet Appendix B.8.

The CRAs may still underinvest or overinvest in information production in the setting with two CRAs, depending on the parameters of the model. The CDS market can still increase surplus from its disciplining effect; if $\alpha_1^{mult} < \hat{\alpha}^{mult}$ and the speculator's cost of information is sufficiently small, an increase in market discipline t —spurred by a decrease in ρ_s or an increase in n —leads to higher surplus, by pushing the equilibrium accuracy α_1^{mult} closer to its optimal level.

The second part of the proposition compares surplus across two different

(b) Proposition 6: two CRAs and surplus

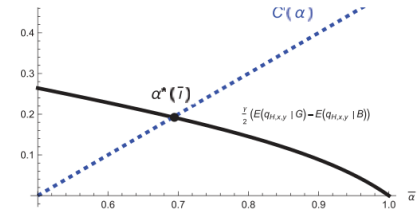


Figure A2
Equilibrium rating accuracy for a given level of market informativeness
As example, $C(\alpha) = (\alpha - \frac{1}{2})^2$, $\bar{t} = \frac{4}{5}$, $q_0 = \frac{1}{2}$, $\gamma = 3$, $f_B = 0.6$, $f_G = 0.2$.

$$= -\frac{1-q_0}{a_0} \sum \frac{(q_{H,x,y})^2}{\Pr(x,y|G)} [\Pr(x,y|G) - \Pr(x,y|B)]^2 < 0.$$

(b) Figure A2: rating accuracy best response

16 Notes-created tables

16.1 Key objects in the model

Object	Meaning
$\theta \in \{B, G\}$	Project quality: bad or good.
f_B, f_G	Default or failure probabilities of bad and good projects, with $f_B > f_G$.
$\Delta f = f_B - f_G$	Information sensitivity of the project outcome to project quality.
I	Investment cost.
$v_\theta = 1 - f_\theta - I$	Net present value of a project of quality θ .
$\tilde{r} \in \{H, L\}$	CRA's private credit report offered to the issuer.
$r \in \{\tilde{r}, \emptyset\}$	Public rating outcome observed by investors; $\emptyset$ denotes no public rating.
p^r	Price investors pay for the claim given rating outcome r .
p^{cds}	Price or spread in the CDS market.
$x_n \in \{-n, +n\}$	Noise trader demand in the CDS market.
η	Probability the second CDS trader is an informed speculator.
ι	Speculator's signal precision; the paper's measure of market informativeness.
$K(\iota) = \rho_s k(\iota)$	Cost of speculator information acquisition.
α	Opportunistic CRA's rating accuracy.
$C(\alpha) = \rho_{cra} c(\alpha)$	Cost of CRA information acquisition.
γ	Weight on the CRA's reputational payoff.
ϕ	Fraction of issuer surplus captured by the CRA as rating fee.

16.2 Main results

Result	Formal message	Intuition
Lemma 3	$\iota^*(\alpha)$ is unique and decreases with expected rating accuracy.	Accurate ratings reduce mispricing and therefore reduce the speculator's value of information.
Lemma 4	$\alpha^*(\iota)$ is unique and increases with market informativeness.	Informative CDS prices make rating errors more visible, increasing CRA reputational discipline.
Proposition 1	There is a unique joint equilibrium (α^*, ι^*) .	Upward-sloping CRA response and downward-sloping speculator response cross once.
Lemma 5	Higher noise-trader volume raises both ι^* and α^* ; stronger CRA reputation raises α^* but lowers ι^* .	Market depth encourages informed trading, while direct CRA discipline crowds out mispricing-based information acquisition.
Corollary 1	The model remains unique with two rating markets.	The disciplining logic survives a dynamic rating-market extension.
Proposition 2	Market discipline increases surplus when CRA accuracy is too low, but may lower surplus when accuracy is too high.	Information production has both screening benefits and real resource costs.
Proposition 3	Linked primary and secondary markets preserve uniqueness, but raise the threshold for market existence.	Investors price in future liquidity costs from trading against an informed speculator.
Lemma 8	With two CRAs, the project is undertaken only when both ratings are high.	Split ratings are not purchased or publicized because they do not support financing.
Proposition 4	The CDS market still disciplines CRAs with multiple CRAs.	CRA competition is not the same as market discipline and can weaken incentives.
Proposition 5	With rating inflation, the CRA may mix between information acquisition and uninformed high ratings.	More market informativeness makes inflated ratings easier to expose.
Lemma 10	More informative project outcomes raise rating accuracy but have ambiguous effects on market informativeness.	Defaults discipline CRAs, but they may arrive too slowly to substitute for price signals.
Proposition 6	Two CRAs without CDS information can produce more or less surplus than one CRA with CDS discipline.	Extra rating agencies and credit-risk market information are distinct governance mechanisms.

16.3 Policy interpretation

Policy or market change	Predicted direct effect	Important caveat
Increase CDS market liquidity	Raises speculator incentives and market informativeness.	Helpful when CRA underinvests in accuracy; potentially wasteful when CRA overinvests.
Lower informed trading costs	Raises ι^* and can raise α^* .	The benefit depends on whether extra CRA accuracy improves surplus enough to offset information costs.
Increase CRA reputational sanctions	Raises α^* directly.	Can reduce ι^* because better ratings leave less mispricing for speculators.
Encourage additional CRAs	Adds another source of screening.	Not necessarily superior to market discipline because issuers can selectively publicize favorable ratings.
Improve disclosure in credit-risk markets	Can make prices more informative and rating errors more transparent.	Requires prices to be linked to credit risk and observable by investors.
Rely on defaults/outcomes	Outcomes eventually reveal whether the rating was accurate.	Defaults are delayed and noisy, so they may not provide timely interim discipline.
Use broad financial market prices	May give macro or equity information.	The discipline is specific to markets whose prices reveal credit-risk information relevant to the rating.

17 Short summary

- A CRA rating is followed by a market for credit risk.
- The CDS price is informative because a speculator may acquire information and trade against noise traders.
- More informative trading makes rating mistakes more transparent.
- This raises the CRA's reputational incentive to acquire accurate information.
- But more accurate ratings reduce mispricing and reduce the speculator's incentive to acquire information.
- The CRA and speculator best responses jointly determine rating accuracy and market informativeness.
- Credit-risk market depth can crowd in rating accuracy.
- Direct CRA regulation can crowd out market informativeness.
- Market discipline improves welfare when the CRA underinvests in accuracy, but can reduce welfare when the CRA overinvests.

- Multiple CRAs, project outcomes, and media are useful sources of discipline, but do not eliminate the special disciplining role of credit-risk prices.

18 Conclusion

18.1 Core conclusion

Credit-risk markets can discipline CRAs by making rating errors more visible. The discipline operates through reputation: if the market price reveals that a rating was likely wrong, investors downgrade their beliefs about the CRA's type, reducing its future payoff.

18.2 Economic intuition

The model is an equilibrium theory of feedback between an information intermediary and a market price. The CRA affects the market by changing the amount of mispricing available to speculators. The market affects the CRA by changing how easily investors can detect rating mistakes.

18.3 Incremental contribution revisited

The paper's main contribution is not simply that market prices contain information. It is that the informativeness of those prices is endogenous to the behavior of the intermediary being disciplined. This creates a subtle interaction between regulation, market design, rating accuracy, and welfare.

18.4 Takeaway

Market discipline is strongest when the market price is informative, timely, and directly connected to the credit risk being rated. Policies that foster such markets can improve rating accuracy, but they must be evaluated jointly with direct CRA regulation and the social cost of information production.